

DOC-4 — System Backup & Recovery Policy

AegisCare HMS / LabTrack v5.5

System	AegisCare HMS / LabTrack v5.5
Supabase project	tgynjasgnerucrlwedui
Hosting	Lovable managed hosting (https://aegiscarehms.lovable.app)
Document status	DRAFT for certification submission
Date	2026-08-12
Related documents	DOC-1 (Architecture), DOC-6 (Security), DOC-7 (Gap Analysis)

Section 1 — Backup Strategy

1.1 Database. The PostgreSQL database is managed by Supabase Cloud, which provides platform-managed backups:

Tier	Provision
Current hosting (Lovable-managed project, free tier)	Daily backups with 7-day retention (Supabase standard for free-tier projects); no point-in-time recovery (PITR) on this tier
Recommended (post-transfer to facility-owned Supabase project)	Enable PITR (up to 7-day granular recovery) and extend backup retention per facility policy

1.2 Schema. The complete authoritative schema lives in version control: 85 migration files in supabase/migrations/ (applied in filename order). The schema can be fully reconstructed on a fresh Supabase project by replaying the migrations (docs/facility-onboarding.md, Step 3). This makes the repository the primary schema-backup and schema-recovery artefact.

1.3 Code. Application source (routes, components, edge functions, config) is versioned in the repository (fmurage6331-dev/confit-core); edge-function code is redeployable from supabase/functions/ (Section 6).

1.4 Secrets. Edge-function secrets and environment variables are stored in Supabase (Edge Functions → Secrets) and the Lovable environment (VITE_SUPABASE_URL, VITE_SUPABASE_PUBLISHABLE_KEY). ☐ PENDING: a facility-held secrets inventory (copy kept offline in the facility safe) is recommended; no secrets are stored in the repository.

1.5 Verification of backups. Supabase backups are verified by restoration drill per Section 8.

Section 2 — Recovery Time Objective (RTO)

2.1 Target RTO (clinical availability):

- 4 hours for the application to be serving again after a data-loss or hosting incident (restore path: Lovable redeploy + Supabase restore + migrate replay).
- 24 hours for full verification of data completeness after recovery (data-integrity checks per Section 8).

2.2 Rationale: the deployment is a single-facility Model A deployment; a 4-hour target is achievable with the migration-file schema path plus Supabase restore, and is consistent with continuity expectations for a Level FACILITY_LEVEL facility. GAP: an actual timed restore drill has not yet been performed —

scheduled the first drill before go-live (Section 8)

DOC-2 — certification documents

Section 3 — Recovery Point Objective (RPO)

3.1 Target RPO: ≤ 24 hours (free tier daily backups) — i.e., at most one day of data can be lost in a worst-case restore.

3.2 Post-transfer improvement: with PITR enabled on the facility-owned Supabase project, RPO improves to ≤ 5 minutes (or per the selected PITR retention window).

3.3 Business expectation: because encounters are closed (signed) daily and MOH reporting is monthly, a 24-hour RPO is acceptable for the current phase;

□ PENDING: confirm with the facility management and record the accepted RPO in the facility business-continuity plan.

Section 4 — Migration Files as Schema Backup

4.1 The 85 migration files in `supabase/migrations/` are the authoritative, version-controlled definition of every table, view, type, function, trigger, policy, seed and cron job:

Migration contents (representative) group

Core schema: `lab_tests`, `roles`, `app_settings`, `test_templates`, `insurance_providers`, `rooms/kinds`, `RLS policies`, `patient_registrations view` + `INSTEAD OF triggers`, `audit_log`, `stock`, `MOH engine`, `wards/beds`

20260716*

Full enumerated schema + routing/billing functions + `encounter_diagnoses`, `icd11_codes`, `views`

20260716*

(consolidated)

MOH 705 mappings/report, dashboard stats, emergency referral tracking, laboratory orders

20260721* (`lab_orders/lab_results`), prescription display names, MOH production sync

20260722*

Radiology routing, ICD-11 encounter-diagnoses sync, DHA task D security roles (19-role is `is_approved`), task E audit completeness (20 tables), `profiles table`

20260730*

Sprint 5.2/6 foundation: `sha_fund_type`, `sha_tariffs` (Legal Notices 146/147), `generate_fhir_encounter()`, `consent_otps`, audit hardening, SHR foundation

Break-glass, SHR access notification, `shr_transmission_log`, Second Schedule FHIR fields, `RLS missing tables`, insurance room kind, cron `accrue-daily-bed-charges`, audit archiving (`archive_old_audit_logs` + cron), contracted prices, inpatient orders, room-kind unification, mortuary, MCH/FP indicators

NLMIS code, MedicationDispense FHIR, encounter signing/locking, dispense stock guard, lab specimen/critical, MAR, mortuary external billing, security hardening, insurance rules, cron `accrue-daily-mortuary-charges`, insurance visit-limit trigger, drift fixes (`encounters/patients/app_settings` SHA columns, `dha_outbound_queue`, `consent_otps`, council columns), SHA benefit packages, claims tables, auto-claim trigger

4.2 Recovery procedure (schema):

```
# 1. Create a new Supabase project
# 2. Run every migration file in filename order (oldest first) in the SQL editor
#    (or via supabase CLI: supabase db push)
# 3. Re-apply grants/secrets, then verify with Section 8 checks
```

4.3 All migrations are written with IF NOT EXISTS / OR REPLACE / ON CONFLICT DO NOTHING guards so replay is safe (verified during the 2026-08-12 transfer-safety drift fixes, 20260812000003-20260812000008).

Section 5 — Data Export Procedures

5.1 Database dump (routine/export path): Lovable dashboard → Settings → Database dump — produces a full PostgreSQL dump of the project (tgynjasgnerucrlwedu). The dump is the primary data recovery artefact (complements the migration-based schema recovery).

5.2 Alternative: supabase db dump via the Supabase CLI (requires project access credentials held by the administrator).

5.3 Frequency: take a database dump:

- immediately before any significant migration or configuration change;
- at least weekly during the certification phase;
- monthly thereafter, retained for 13 months (aligns with audit retention);
- the dump must be stored off-platform (facility-controlled storage, encrypted) — $\triangle$ GAP: designate the storage location (FACILITY_BACKUP_STORE) and confirm the encryption-at-rest posture.

5.4 Patient data export (right of access): for a data-subject access request, the facility uses the same dump path or the on-demand FHIR resources (fhir-patient, fhir-encounter, fhir-condition edge functions) to assemble the record (DOC-3 §7).

Section 6 — Edge Function Recovery

6.1 The five edge functions (claims-dispatcher, fhir-patient, fhir-encounter, fhir-condition, icd11-search) are source-controlled in supabase/functions/ and redeployed with:

```
supabase functions deploy claims-dispatcher --project-ref tgynjasgnerucrlwedu
supabase functions deploy fhir-patient      --project-ref tgynjasgnerucrlwedu
supabase functions deploy fhir-encounter    --project-ref tgynjasgnerucrlwedu
supabase functions deploy fhir-condition    --project-ref tgynjasgnerucrlwedu
supabase functions deploy icd11-search      --project-ref tgynjasgnerucrlwedu
```

6.2 Automated deployment also exists via GitHub Actions (.github/workflows/deploy-functions.yml, triggered on main pushes touching supabase/functions/).

6.3 Secrets to re-apply after a project rebuild: ICD_CLIENT_ID, ICD_CLIENT_SECRET, plus the future integration secrets (DHA/SHA/AfyaLink/HWR/Africa's Talking) defined in docs/integration-activation-manual.md.

6.4 JWT verification config (supabase/config.toml: verify_jwt = true for all five functions) must be reapplied with deployment.

Section 7 — Cron Job Recovery

Three pg_cron jobs are defined in migrations; after a schema rebuild they are restored by re-running the defining migrations, or manually:

Job name	ScheduleEAT (UTC)	Function	Defining migration
accrue-daily-bed-charges	1 21 * * 00:01 (21:01)	accrue_daily_bed_charges() — inserts daily ward bed-charge line items for admitted admissions	20260806000007_accrue_bed_charges
accrue-daily-mortuary-charges	31 21 * * 00:31 (21:31)	accrue_daily_mortuary_charges() — accrues storage charges for mortuary_records with status='stored';	20260807000004_sprint10b_mortuary.s
archive-audit-logs-nightly	0 23 * * 02:00 (23:00)	archive_old_audit_logs() — moves audit rows older than 2 years to audit_log_archive, logs runs in audit_archive_runs	20260806101050_e8638a1f-75b6-42c3

Manual re-registration pattern (idempotent):

```
SELECT cron.unschedule('accrue-daily-bed-charges')
WHERE EXISTS (SELECT 1 FROM cron.job WHERE jobname = 'accrue-daily-bed-charges');
SELECT cron.schedule('accrue-daily-bed-charges', '1 21 * * *',
    $$ SELECT public.accrue_daily_bed_charges(); $$);
```

Requires the pg_cron and pg_net extensions (installed by 20260806101050...).

Section 8 — Testing & Verification

8.1 Post-restore verification checklist (run in the restored project's SQL editor):

```
-- 1. Schema completeness: core tables exist
SELECT count(*) FROM information_schema.tables
WHERE table_schema = 'public'; -- expect the full table set

-- 2. Security hardening state (fails loudly if permissive policies reappeared)
-- (mirrors the DO block in 20260811000007_security_hardening.sql)

-- 3. Audit pipeline works
SELECT public.archive_old_audit_logs();
SELECT * FROM public.audit_archive_runs ORDER BY run_at DESC LIMIT 5;

-- 4. FHIR generator works (sample output for certification)
SELECT generate_fhir_encounter('<encounter-uuid>');

-- 5. SHA claim auto-generation works on a signed insurance encounter
SELECT id, claim_number, status FROM public.sha_claims
WHERE encounter_id = '<signed-insurance-encounter-uuid>';

-- 6. Cron jobs registered
SELECT jobname, schedule FROM cron.job;
-- expect: accrue-daily-bed-charges, accrue-daily-mortuary-charges,
--         archive-audit-logs-nightly

-- 7. RLS gates work
SELECT public.is_approved(auth.uid());
```

8.2 Application smoke test: login as admin → open Dashboard, Queue, a room (consultation), Laboratory, Accounting, Admin → Audit Log, Admin → Claims Queue (admin.queue.tsx) and confirm data renders.

8.3 Restore drill:  GAP — schedule and document the first full restore drill (schema replay + data dump restore + verification) before go-live, and annually thereafter. Record drill dates/results in the facility continuity file.

Section 9 — Disaster Recovery Procedure

9.1 Invocation. This procedure is invoked on: total data loss; unrecoverable project corruption; extended platform outage; or regulatory requirement to restore.

9.2 Team. Facility administrator + system developer; Lovable/Supabase support as escalation.

9.3 Step-by-step:

1. Assess — determine scope (data only / schema only / both), confirm the latest database dump and its timestamp (RPO check).
2. Create replacement Supabase project — same region where feasible; note the new project ref.
3. Replay schema — run all 85 migration files in order (Section 4).
4. Restore data — apply the database dump (Section 5); where the dump predates the latest migrations, replay the remaining migrations after the dump (migrations are replay-safe).
5. Reconfigure — environment variables (VITE_SUPABASE_URL, VITE_SUPABASE_PUBLISHABLE_KEY), edge-function secrets (Section 6.3), Auth settings, storage buckets.
6. Redeploy edge functions (Section 6) and verify verify_jwt config.
7. Re-register cron jobs (Section 7).
8. Verify — run Section 8 checks; confirm audit continuity (audit_archive_runs), FHIR generation, claims tables, and RLS.
9. Point the app at the new project — update Lovable environment and redeploy; update DNS if the URL changed.
10. Notify — inform staff, DHA/SHA contacts if integrations were live, and affected parties per DOC-6 §9.

9.4 Authoritative DR reference — the 13-chapter self-hosted transfer manual. The complete facility-transfer and disaster-recovery runbook is documented in the 13-chapter self-hosted transfer manual produced by the AegisCare development team (repository handoff documentation). Chapters cover: repository forking and clone; Supabase project creation; migration replay; Lovable connection; facility configuration; secrets; admin user creation; verification checklist; edge function deployment; cron restoration; data export/dump; cut-over and rollback; and post-go-live monitoring. $\triangle$ GAP: that manual is not present inside this checkout (it is a handoff artefact) — attach the current copy to this certification pack and keep it versioned alongside DOC-1.

9.5 Rollback path: migrations are replay-safe and additive; rollback of a failed change is by corrective migration (never by editing history) and a pre-change database dump per Section 5.3.

Approval

Role	Name	Signature	Date
Prepared by	(Technical Lead)		
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Approved by	(Management)		

